# TIC 257806221

Eclipsing binary candidate dossier - Star Vetter

**PERIOD ALIAS**

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BLS EPHEMERIS
  Period       9.962056 d  (ALIASED, see below)
    -> corrected period: 19.924111 d
  Epoch (t0)   2800.0852 BTJD
  Duration     0.2440 d
  Depth        5.250 %  (52,502 ppm)
  SNR          161.7

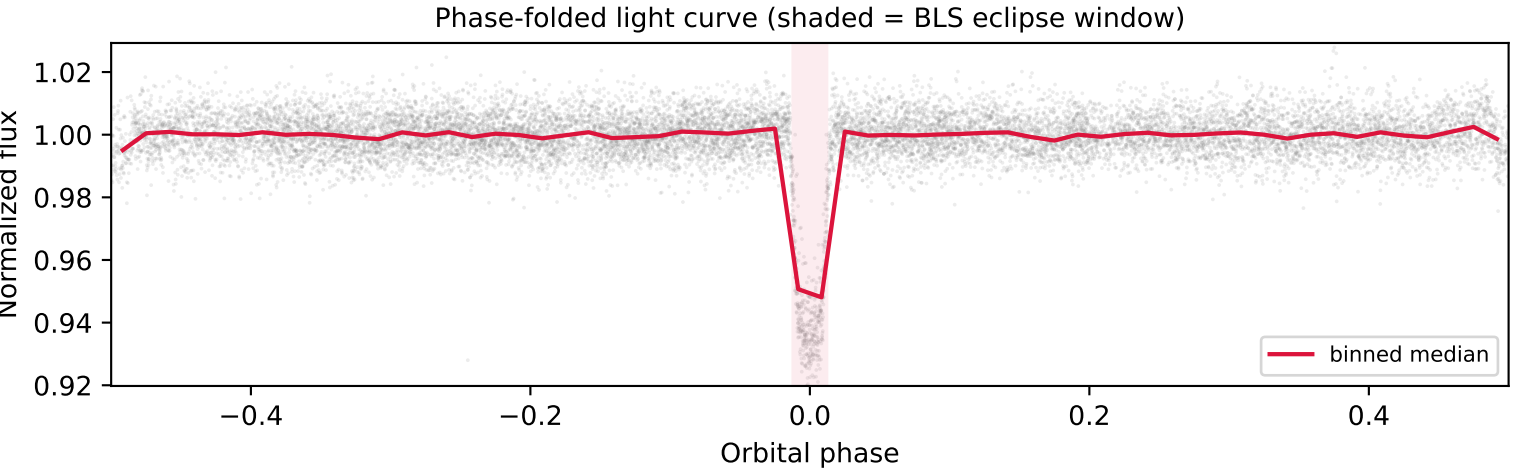
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     6  (gate >= 3)
  Robust depth ratio    1.10  (gate >= 0.50)
  Robust depth sigma    143.7  (gate >= 7.0)
  Odd/even z-score      3.33  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

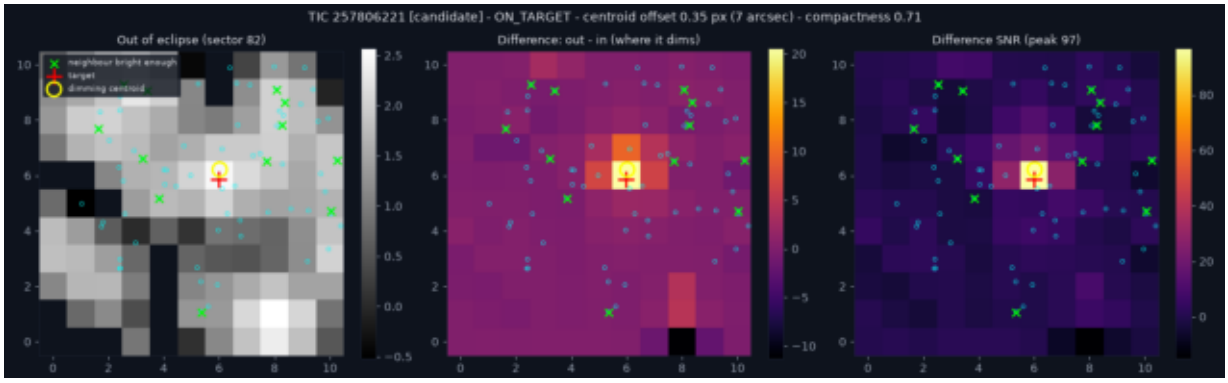
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 82    light-curve dip in this sector: 113.8 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   96.86   (gate >= 5.0)      Compactness   0.71   (gate >= 0.50)
  Dimming centroid offset     0.35 px (7 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 257806208
    1.72 px away, dTmag +1.08 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.33 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 19.924111 d, in any write-up.